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Abstract

Purpose: The purpose of this paper is to propose a new methodology that handles the issue of the dynamic behavior of customers over time.

Design/methodology/approach: A new methodology is presented based on time series clustering to extract dominant behavioral patterns of customers over time. This methodology is implemented using bank customers' transactions data which are in the form of time series data. The data include the recency (R), frequency (F) and monetary (M) attributes of businesses that are using the point-of-sale (POS) data of a bank. This data was obtained from the data analysis department of the bank.

Finding: After carrying out an empirical study on the acquired transaction data of 2531 business customers that are using POS devices of the bank, the dominant trends of behavior are discovered using the proposed methodology. The obtained trends were analyzed from the marketing viewpoint. Based on the analysis of the monetary attribute, customers were divided into 4 main segments including high-value growing customers, middle-value growing customers, prone to churn, and churners. For each resulted group of customers with a distinctive trend, effective and practical marketing recommendations were devised to improve the bank relationship with that group. The prone to churn segment contains most of the customers; therefore, the bank should conduct interesting promotions in order to retain this segment.

Practical implications: The discovered trends of customer behavior and proposed marketing recommendations can be helpful for banks in devising segment-specific marketing strategies as they illustrate the dynamic behavior of customers over time. The obtained trends are visualized so that they can be easily interpreted and utilized by banks. This paper contributes to the literature on customer relationship management (CRM) as the proposed methodology can be effectively employed on different businesses to reveal trends in customer behavior.

Originality/value: As in the current business condition, customers' behaviors are changing continually over time and the switching cost of customers has reduced significantly, choosing an effective customer segmentation methodology which can consider the dynamic behaviors of customers is essential for every business. This paper proposes a new methodology to capture customer dynamic behavior using time series clustering on time-ordered data. This is an improvement over previous studies, in which static segmentation approaches have often been adopted. This is the first study that combines the RFM model and time series clustering to reveal trends in customer behavior.

Keywords: Customer relationship management (CRM), segmentation, customer behavior, time series, time series clustering, similarity measures

1. Introduction

CRM (Das and Mishra, 2019, Li et al., 2018, Kumar and Reinartz, 2018, Rahimi et al., 2017) has gained increased attention during the recent years. In addition, CRM has been the main focus of firms that are active in highly competitive industries including banking, telecommunication, etc. Ngai et al. (2009) categorized the CRM framework into operational and analytical CRM. The main focus of the operational CRM is automation of the customer-centered business process while the analytical CRM mainly concentrates on the analysis of customers' behaviors in order to assist companies in customer management decisions. Data mining is defined as a process of extracting and discovering hidden patterns from large data (Han et al., 2011). Data mining tools have been widely used in the context of analytical CRM (Lin et al., 2017, Shmueli et al., 2017). Clustering is an unsupervised data mining technique which is extensively utilized for customer segmentation. Particularly, customer segmentation is a popular approach which attempts to distinguish between different groups of customers. Customer segmentation is an important CRM activity which helps managers to gain a better understanding of customers' needs and behaviors (Tsiptsis and Chorianopoulos, 2010). There are many approaches for customer segmentation, among which value-based segmentation (Parvaneh et al., 2012, Khajvand and Tarokh, 2011, Kim et al., 2006) has been the focus of various studies. In addition, in the context of value-based segmentation, the RFM model developed by (Hughes, 2000) was employed in many researches. Reviewing the literature on RFM-based customer segmentation indicates that most of the studies adopted a static segmentation approach. One of the most crucial issues in customer segmentation is emphasizing "customer dynamics" (Akhondzadeh-Noughabi and Albadvi, 2015). In fact, due to the impact of psychosocial and environmental determinants, including changes of market, promotion, new product introduction and competitors' strategies, customer behavior is changing over time (Akhondzadeh-Noughabi and Albadvi, 2015). Thus, to be effective, a customer segmentation process must consider the dynamic nature of customer behavior (Chong and Chen, 2010, Böttcher et al., 2009, Ha, 2007, Ha and Bae, 2006, Kim et al., 2005, Chen et al., 2005, Song et al., 2001).

As pointed out by Akhondzadeh-Noughabi and Albadvi (2015), the main drawback of static segmentation models is that they assume customer segments and their members as stable; however, in the real world, customer behavior is not stable and evolves over time. Furthermore, models achieved utilizing static segmentation approaches are valid only during a particular time point. Therefore, it is not enough to adopt a static customer segmentation approach as it leads to missing important trends and patterns of customer behavior. In addition, without extracting and discovering trends of customer behavior, we cannot predict the future behavior of customers accurately. Involving time in analysis is one of the main approaches to dynamic customer segmentation. The sequential rule mining and time series analysis are two main techniques which are used in this area. Most of works focused on sequential rule mining (e.g., Rahimi et al., 2017(Akhondzadeh-Noughabi and Albadvi, 2015, Böttcher et al., 2009, Song et al., 2001)) and there are few papers based on time series clustering in the customer segmentation concept (e.g., (Saas et al., 2016,

Menéndez et al., 2014). Moreover, there are two studies that have considered the RFM model on various time points (Song et al., 2017, Hosseini and Shabani, 2015). Although a time series of RFM attributes was constructed in these studies, the trend extraction method did not consider the order of data. This problem may cause imperfect results. To tackle this problem, in this study, we propose a methodology to use time series clustering which has the ability to discover hidden patterns in time ordered data effectively.

The input to this methodology is the data of customer activity in terms of RFM attributes in defined time points. Contrary to static segmentation which performs clustering on non-time-oriented data, in this paper, we have gathered time series data and employed time series clustering to visualize dominant trends in customers' behavior. By doing so, we can determine the differentiating characteristics of customer segments. In fact, the main contribution of this study is to consider the trend of changes over time for the RFM model. The proposed methodology consists of three main phases including preprocessing, time series clustering and analysis of results. In the first phase, the task of cleaning and preprocessing data is performed. The result of this phase is the time series data of customer behavior. In the second phase, the task of clustering time series is conducted using different similarity measures such Euclidean distance, correlation-based measure (COR), dynamic time warping (DTW) and temporal correlation coefficient (CORT) (Montero and Vilar, 2014); the best clustering result is obtained by computing the silhouette index (Desgraupes, 2013), which is a widely-used cluster validity index. Finally, in the analysis phase, the result of clustering is explored and the dominant trends in customer behaviors are extracted. A case study using a bank's data is conducted, and the usefulness of the proposed methodology is tested successfully.

The rest of the paper is as follows: in section 2, we give some background on CRM, customer segmentation using RFM, customer trend analysis, and time series clustering. Section 3 describes the proposed methodology in details. In section 4, we present the results of an empirical study and analyze the customer trends to make managerial recommendations about how to develop relationship strategies to deal with each customer group. Section 5 concludes the paper and discusses future work arising from this study.

2. Literature review

In this section, we describe some concepts and methods utilized throughout this paper.

2.1 Customer relationship management (CRM)

CRM is an essential supporting element of each business strategy. CRM consists of a set of processes and enabling systems supporting a firm strategy to build long-term customer relationship (ling & yen 2001). Although CRM is an important business concept, there is still no universally accepted definition of CRM. However, the intersection of all definitions is that CRM is a comprehensive process of acquiring and retaining customers by utilizing business intelligence techniques to maximize the value of both firms and customers (Ngai et al., 2009). According to some researches (Ngai et al., 2009, Kracklauer et al., 2004, Swift, 2001, Parvatiyar and Sheth,

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2001), CRM is described as a four-phase cycle including customer identification, attraction, retention, and development.

Ngai et al. (2009) provided an excellent correspondence among the four-phase cycle and data mining techniques. There is another categorization on CRM, which has divided CRM activities into two main approaches consisting of operational CRM and analytical CRM (Ngai et al., 2009).

In fact, the scope of this paper is the analytical CRM. The main purpose of the analytical CRM is to analyze customers' behavior to distinguish customers and develop different CRM strategies for each type of customer group (Ngai et al., 2009). In other words, the main focus of the analytical CRM is to obtain analytical information from customer data (Jayanthi and Vishal, 2011). In the area of the analytical CRM, data mining tools and techniques are a widely-used means of customer data analysis. Association, classification, clustering, forecasting, regression, sequence discovery and visualization are the seven general data mining models that have been employed in CRM (Ngai et al., 2009). The four-phase CRM cycle begins with customer identification that includes customer segmentation and target customer analysis. Customer segmentation involves grouping customers based on their characteristics and target customer analysis involves discovering profitable customer segments (Ngai et al., 2009, Woo et al., 2005). Therefore, customer segmentation is a significant step in the CRM process.

2.2 Customer segmentation using RFM

Segmentation is the process of subdivision of customers into distinct and meaningful customer groups based on customers' features (Desgraupes, 2013, Tsiptsis and Chorianopoulos, 2010, Ngai et al., 2009). The main benefit of segmentation is to help firms to distinguish customers and develop efficient segment-specific strategies according to characteristics of each segment (Tsiptsis and Chorianopoulos, 2010, Ngai et al., 2009). In the business-to-customer (B2C), customer segmentation approaches are categorized into customer need-based, characteristics-based (Greengrove, 2002), and value-based segmentations (Kim et al., 2006). In recent years, there is a large research stream that has utilized value-based segmentation. One of the value-based segmentation models is the RFM model which has been widely-applied for customer value analysis.

There are many models introduced for measuring customer value, among which RFM developed by Hughes (2000) is a popular and widely-used model. A body of researches have performed segmentation of customers using RFM variables. Table 1 shows some important researches in this area. Although many researches have adopted the RFM model for customer segmentation, most of the researches are based on the theory of static segmentation. In fact, there are few researches that have utilized dynamic segmentation approaches (e.g., researches in Table 1 denoted by D in the static /dynamic segmentation). The main drawback of the static segmentation models is that these models assume that customer segments and their members are stable; however, in real-world, customer behavior is not stable and evolves over time (Akhondzadeh-Noughabi and Albadvi, 2015, Böttcher et al., 2009). As mentioned previously (Akhondzadeh-Noughabi and Albadvi,

2015), knowledge and insight gained by exploiting the static models are valid only for a particular time point and do not reflect the complex and evolving nature of customer behavior. Employing static segmentation approaches leads to loss of important trends and patterns about customers' behavior (Akhondzadeh-Noughabi and Albadvi, 2015, Böttcher et al., 2009, Ha and Bae, 2006, Ha et al., 2002).

2.3 Customer trend analysis

There are many researches in the context of customer trend analyzing. Reviewing the literature about customer trend analysis indicates that the main focuses of these researches are as follows:

(a) Exploiting the Markov chain process to model the customer transition between segments (e.g., (Mark et al., 2013, Lemmens et al., 2012, Netzer et al., 2008, Ha et al., 2002)). These researches utilized different Markov chain models to articulate the transition probabilities of customers between segments.

(b) Discovering the principal patterns of customer movements (e.g., (Akhondzadeh-Noughabi and Albadvi, 2015, Hosseini and Shabani, 2015, Ha, 2007, Ha and Bae, 2006)). This category has a close relationship with our study. The main aim of these studies is to extract movement patterns that customers experience over time.

(c) Investigation of determinants that may impact customer shifting behaviors between segments (e.g., (Kim and Bell, 2011, Lemon and Mark, 2006, Brangule-Vlagsma et al., 2002)). This category of researches which is an interesting area of research attempts to explore and identify factors influencing customer behavior to move between segments.

According to the above categorizations of customer trend analysis works, our proposed method relates to tracking customer shifts between segments and specifically to the second category named as discovering the principal patterns of customer movements as it attempts to extract dominant trends of customer behaviors over time. To be more specific, we have accomplished the task of analyzing customer trends using time series analysis and particularly time series clustering. In the rest of this subsection, we describe important issues in this regard.

2.4 Time series analysis

Time series clustering and forecasting techniques are broadly employed in many areas of applications. In the forecasting category, the autoregressive integrated moving average model (ARIMA) (Box et al., 2015, Brockwell et al., 2002) method and the moving average (Yager, 2008) are the standard and widely-used techniques.

During the recent years, some forecasting techniques based on the ordered weighted averaging (OWA) operator have been proposed such as the generalized ordered weighted moving average (GOWMA) (Merigo and Yager, 2013), induced GOWMA (IGOWMA) operator (Merigo and Yager, 2013) and heavy ordered weighted moving average (HOWMA) operator (Avilés-Ochoa et al., 2018). The description of the time series clustering techniques is given in the following.

2.4.1 Time series clustering

For clustering time series, a similarity measure along with a clustering algorithm must be chosen. Several similarity measures have been proposed in the literature (Aghabozorgi et al., 2015). As stressed in the literature, there is no similarity measure that performs well in all contexts. Therefore, based on the context and nature of data of time series, an appropriate similarity measure should be selected. In the following, we briefly describe four popular similarity measures including Euclidean distance, COR, CORT, and DTW (Montero and Vilar, 2014) that have been employed for time series clustering in various articles.

2.4.2 Similarity measures

Given two time series $S = s_1, s_2, s_3, \ldots, s_n$ and $Q = q_1, q_2, q_3, \ldots, q_n$, in the following, we describe the utilized similarity measures.

2.4.2.1 Euclidean distance

The Euclidean distance between the two time series S and Q is defined as (Montero and Vilar, 2014):

$$d_{L2}(S,Q) = \left(\sum_{t=1}^{n} (S_t - Q_t)^2 \right)^{\frac{1}{2}} \quad (1)$$

2.4.2.2 Dynamic time warping (DTW)

DTW is based on computing the alignment between two time series with arbitrary length by employing a dynamic programming approach. This algorithm was firstly applied in speech recognition (Sakoe and Chiba, 1978). It associates two time series through warping the time axis of one series onto another.

Considering the two time series of S and Q, the DTW distance can be described by the equation (2):

$$DTW(S,Q) = \min_{r \in M} \left(\sum_{m=1}^{M} \left| s_{im} - q_{jm} \right| \right), \quad (2)$$

where the path element $r = (i, j)$ represents the relationship between the two series. The goal is to find the optimal warping path. The optimal path characterizes the path that minimizes the overall distance between S and Q (Al-Naymat et al., 2009). As DTW is based on a dynamic programming paradigm, this method is expensive in computation (Aghabozorgi et al., 2015).

2.4.2.3 Correlation-based measure (COR)

This measure computes dissimilarity between two time series using Pearson's correlation factor. For the two time series S and Q, the COR measure is formulated by the following relation (Equation (3)):

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$$COR(S,Q)=\frac{\sum_{t=1}^{n}\left(S_t-\bar{S}\right)\left(Q_t-\bar{Q}\right)}{\sqrt{\sum_{t=1}^{n}\left(S_t-\bar{S}\right)^2}\sqrt{\sum_{t=1}^{n}\left(Q_t-\bar{Q}\right)^2}}, \quad (3)$$

With $\bar{S}$ and $\bar{Q}$ as the average value of S and Q, respectively (Montero and Vilar, 2014).

2.4.2.4 Temporal correlation coefficient (CORT)

This measure considers proximity on both values and behavior when computing the similarity between two time series (Montero and Vilar, 2014, Chouakria and Nagabhushan, 2007). It is defined as the equation (4).

$$CORT(S,Q)=\frac{\sum_{t=1}^{n-1}\left(S_{t+1}-S_t\right)\left(Q_{t+1}-Q_t\right)}{\sqrt{\sum_{t=1}^{n-1}\left(S_{t+1}-S_t\right)^2}\sqrt{\sum_{t=1}^{n-1}\left(Q_{t+1}-Q_t\right)^2}}, \quad (4)$$

2.4.3 Clustering algorithms

Clustering is the task of dividing data objects into groups such that objects in each group are more similar to objects of that group (Han et al., 2011, Tan, 2006). In addition, objects of each group are more dissimilar to objects of other groups. Clustering is an unsupervised data mining technique that has been employed widely in many areas of research (Han et al., 2011). There are many clustering algorithms in the literature which are generally divided into four categories including partitioning, hierarchical, density-based and grid-based methods (Han et al., 2011). Partitioning and hierarchical clustering methods are popular methods in the literature. The K-means algorithm from the partitioning category is a well-known clustering algorithm (as observed in Table 1). Despite successful application in clustering, k-means is not a suitable choice for clustering of time series data. Since it performs clustering based on the computation of similarity of each time series to centroids point of a set of time series, which might not be properly defined (Montero and Vilar, 2014). Therefore, we utilized hierarchical clustering algorithms for clustering of time series.

2.4.3.1 Hierarchical clustering

A hierarchical clustering algorithm operates by organizing data objects into a hierarchy of clusters. This grouping of data objects in a hierarchical way is useful for data visualization (Han et al., 2011). Hierarchical clustering methods are divided into two main approaches including agglomerative and divisive (Han et al., 2011). An agglomerative method firstly assigns each object to individual cluster and iteratively merges clusters into bigger clusters until all objects are grouped in a single cluster or other termination conditions are fulfilled (Han et al., 2011).

On the other hand, a divisive method firstly assigns all objects in one cluster which is called the root cluster. Then, it splits the root cluster into several smaller sub-clusters. The cluster division process continues until each cluster at the lowest level contains only a single object (Tsiptsis and Chorianopoulos, 2010).

In this study, we selected the agglomerative clustering approach. Different methods have been developed in the context of agglomerative clustering, but we exploited the Ward method (Murtagh and Legendre, 2014)which is based on a sum-of-squares criterion. This method generates clusters that minimize within-cluster variance (Murtagh and Legendre, 2014, Murtagh and Contreras, 2012).

Table 1: The literature review on RFM analysis

`Reference	Variables	Algorithms	Domain	Contribution	Static (S)/Dynamic (D) segmentation
(Khajvand and Tarokh, 2011)	RFM	K-means, two-step, x-means, autoregressive integrated moving average (ARIMA), Weighted analytic hierarchy process (AHP)	Retail banking scope	Clustering customer based on RFM variables and then ranking clusters using weighted RFM whose weights obtained using AHP. The ARIMA algorithm was exploited to forecast the customer life time value (CLV).	S
(Parvaneh et al., 2012)	Weighted LRFM (Length)	K-means	Business-to-business (B2B), hygienic manufacturer	Integrating the analytical hierarchy process and LRFM (length, recency, frequency, and monetary) for the retailer segmentation.	S
(Chen et al., 2012)	RFM	K-means, rough set, LEM2 (learning from Examples Module, version 2)	Health Care industry	Segmentation of patients to offer differentiated offering and service based on RFM analysis and the rough set theory	S

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(Cho et al., 2013)	RFM	K-means	Ubiquitous commerce (U-Commerce)	Proposing a personalized recommendation system using clustering of item preference based on RFM in U-commerce	S
(Parvaneh et al., 2014)	LRFMP (length, recency, frequency, monetary, and potential value)	K-means	B2B, hygienic manufacturer	Adding P (potential value) dimension to LRFM in order to perform retailer segmentation. Moreover, the P variable is used for constructing a cross-selling index which was used to develop cross-selling strategies.	S
(Hu and Yeh, 2014)	RFM	RFMP-growth: a pattern growth based algorithm	Chain supermarket	Defining the RFM-pattern and developing a new algorithm to discover the set of approximate RFM customer-patterns without customer identification information.	S
(Khobzi et al., 2014)	RF*M* RF1F2M	K-means	Banking: point of sale (POS) transaction of merchants	This study proposed two adapted versions of RFM including RF1F2M and RF*M* for guild segmentation.	S
(Coussement et al., 2014)	RFM	RFM analysis	Direct marketing	Investigation of the impact of the different levels of data inaccuracies on the performance of three customer segmentation techniques including RFM, CHAID decision tree and logistic regression.	S
(You et al., 2015)	RFM	K-means, CHAID, Pareto ratio	B2B: retail customer, Data of six products	Proposing a decision-making framework for utilizing data-mining algorithms offer personalized customer service.	D
(Akhondzadeh-Noughabi and Albadvi, 2015)	RFM	K-means, generalized sequential pattern (GSP)	Mobile phone operators: call detail records/event detail records (CDR/EDR)	Proposing a combined methodology based on cluster analysis and sequential rule mining in order to discover dominant patterns of customer transition between different clusters over time.	D
(Abirami and Pattabiraman, 2016)	RFM	K-means, association rule mining	Consumers of retail stores	Segmentation of customers using RFM and then discovering the purchase pattern of each segment exploiting association rule mining.	S
(Ansari and Riasi, 2016)	demographic, profitability, loan amount,	Two-step clustering algorithm	Banking	Identification of bank customers segments in order to develop effective marketing strategies.	S

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	account related information				
(Dursun and Caber, 2016)	RFM	Self-organizing maps (SOM) and K-means	Hotel, tourism	The aim of this study was to profile hotel customers using RFM and data mining tools.	S
(Alipour et al., 2016)	RFM, Demographic variables	K-means and Kohonen (SOM) algorithms, association rule mining (A-priori, Éclat, and FP-growth rule mining algorithms)	Global pizza restaurants chain	The focus of this study was to find the best methodology for customer segmentation exploiting the RFM concept.	S
(Song et al., 2017)	RFM	K-means	Telecom	Proposing a statistic based method to value latent user through time series segmenting time interval on large-scale telecom data on the Spark platform.	D
(Serhat et al., 2017)	LRFMP (length, recency, frequency, monetary and periodicity)	K-means	Grocery retail industry	Developing an extended LRFM model, namely LRFMP (length, recency, frequency, monetary, and periodicity) by including the periodicity as a new variable. Further, this study formulated the recency variable to reflect the repetitive purchasing.	S
(DOĞAN et al., 2018)	RFM	Two-step cluster analysis and K-means clustering	Sport retail store	Utilizing RFM analysis and clustering techniques in order to perform sports retail store's customers and consequently devising appropriate marketing strategies.	S

3. The proposed methodology

Our proposed model, as you see in Figure 1, is divided into three main phases including preprocessing, time series clustering and analysis of the results. The data collecting, transforming into the RFM model and preparing data for time series analysis are performed in the preprocessing phase. The main phase of the model is time series clustering in which, the trends of customers are being clustered. In the final phase, the results of clustering are analyzed and then some appropriate business suggestions are proposed for customer groups. In the following, each phase of the proposed methodology is described in more details.

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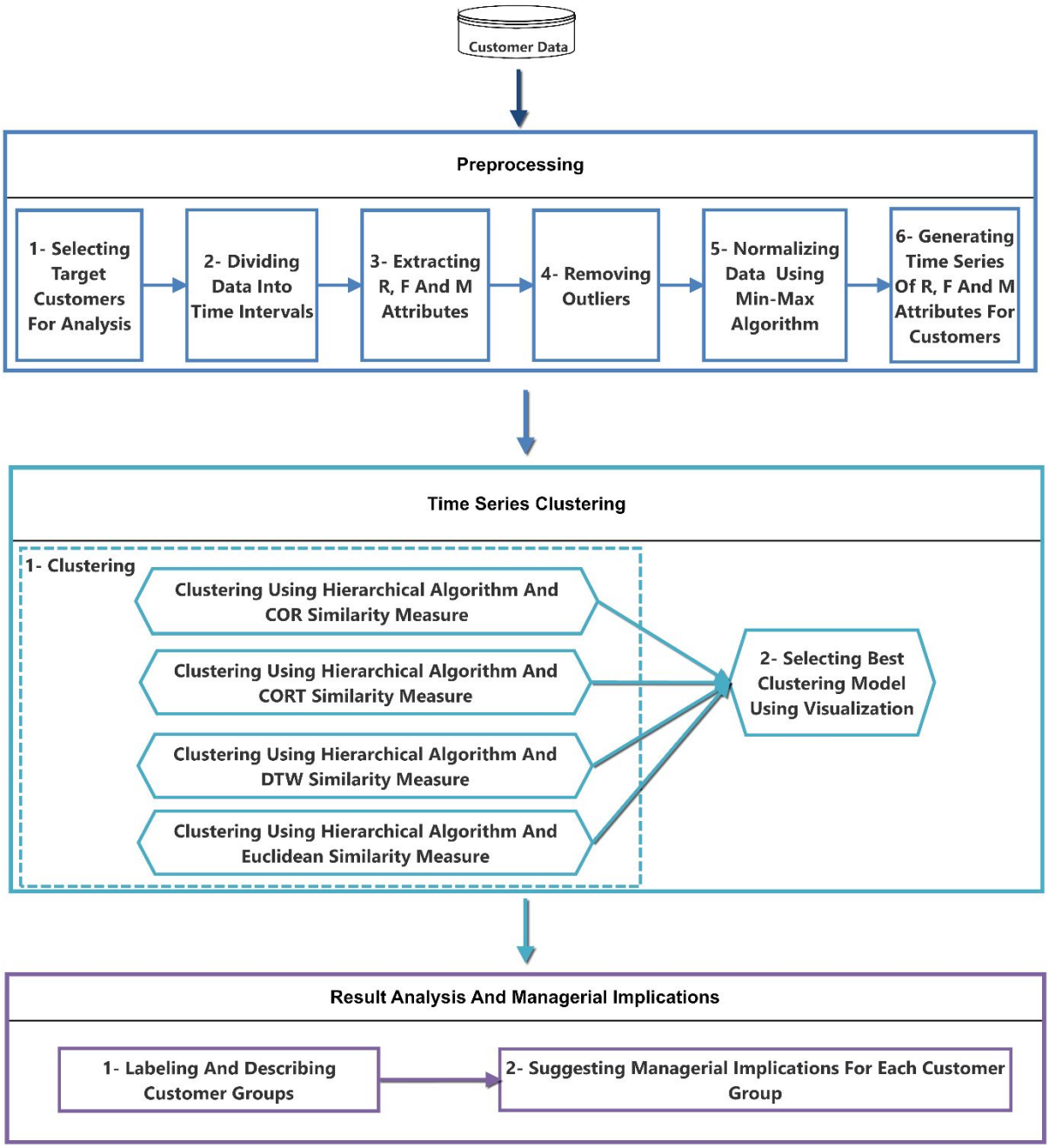


Figure 1: The steps of the proposed methodology

3.1 Preprocessing

In this phase, the data are prepared to perform time series clustering using the following steps:

3.1.1 Selecting target customers

The first step in the analysis is to filter target customers to track their behavior.

3.1.2 Dividing data into time intervals

Given a predefined time interval and considering the time period of study, the time series data is generated.

3.1.3 Extracting the R, F and M attributes

As the model is based on the RFM model, the R, F and M attributes for each customer must be derived from the data. This process must be performed for each time point. According to the RFM definition, the F attribute is the number of customer transactions in a time point. The M attribute is the total amount of purchases in a time point. The R attribute is the days between the last date of purchase and the end of the time point (Hughes, 2000).

3.1.4 Removing outliers

In this step, customers who have missing or incorrect RFM values are removed from data.

3.1.5 Normalizing data using the min-max algorithm

As the range of RFM attributes may differ in each time point, the values of each attribute is normalized using Min-Max algorithm (Han et al., 2011).

3.1.6 Generating time series of the R, F, and M attributes for customers

The final step of the preprocessing phase is to generate the time series of customers' data based on R, F, and M. This data will be used in the time series clustering phase.

3.2 Time series clustering

In this phase, customers are clustered based on their time series data. The steps in this phase are as follows.

3.2.1 Clustering

One key decision regarding time series clustering is the selection of a proper similarity measure. For this purpose, different similarity measures such as Euclidean distance, DTW, COR, and CORT (Montero and Vilar, 2014) are used to find out which one can give more promising results.

3.2.2 Selecting the best clustering model using the cluster validity index

In this step, the results of each clustering with different similarity measure are evaluated using a cluster validity index (e.g., the silhouette index) (Desgraupes, 2013) and the best clustering results are achieved.

3.3 Result analysis and managerial implications

The final phase is to analyze the resulted clusters of the previous phase applying the following steps:

3.3.1 Labeling and describing customer groups

By analyzing the mean trend of each cluster, the behavior of customers belonging to that cluster can be recognized. This analysis is used to label each cluster and to provide description of each cluster.

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3.3.2 Suggesting managerial implications for each customer group

According to the results of analysis in the previous subsection, for each customer group, marketing recommendations are suggested.

4. Empirical study

4.1 Data set

The data which we used in our experiment is POS purchase transactions over a period of seven months, containing over 259000 transactions belonging to 2531 active customers. The fields of raw data can be observed in Table 2.

Table 2: Raw data fields

Field	Description
TerminalID	The ID of the customer.
TransationDate	The date that the transaction occurred.
Amount	The transaction amount.
Install Date	The date that the customer started using the POS device.
PSP Bank	The customer bank.
Terminal Status	The status of using the POS device by the customer.

The steps of implementation of our proposed model are described as follows:

4.2 Preprocessing

According to the proposed methodology, the following steps were accomplished in the preprocessing phase.

4.2.1 Selecting target customers for analysis

Based on our case which is a bank's B2B POS customers, we selected high-value customers. High-value customers are the top ten percent of all customers based on the M and F attributes.

4.2.2 Dividing data into time intervals

We divided our data into seven equal time intervals of 30 days. Each time point contains on average between 37000 and 40000 transactions. The seven-month time period includes April to September.

4.2.3 Extracting the R, F and M attributes

The R, F and M attributes of each customer were extracted for each time point. The R attribute is the days between the date of the last transaction and end of the time point. The F attribute is the number of transactions for each customer at each time point. Moreover, the M attribute is the sum

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of the amount of all transactions in each time point. The sample of the resulted data is illustrated in Table 3.

Table 3: Data with the R, F and M attributes

TerminalID	R	F	M(IRRials)
30123010	15	124	8,241,343
30123012	0	403	31,443,311
87001521	9	203	12,436,574
87002038	0	519	41,452,293

4.2.4 Removing outliers

After the R, F and M attributes for each customer are extracted from the raw data, we removed the outlier data based on simple outlier detection using standard deviation (Kantardzic, 2011). Since all the selected customers have transactions in all the days, the R attribute for all the customers is zero and thus the R attribute was removed from our analysis.

4.2.5 Normalizing data using the min-max algorithm

The normalization of the R, F and M attributes was accomplished independently for all the selected customers. In this study, we employed min-max (Han et al., 2011) normalization.

4.2.6 Generating a time series of the R, F and M attributes for customers

After conducting the described steps, we obtained the time series of the R, F, and M attributes for all the customers. A sample of data is demonstrated in Table 4.

Table 4: Time series data For M attribute

TerminalID	**Month1**	**Month2**	**Month3**	**Month4**	**Month5**	**Month6**	**Month7**
30123010	0.173	0.008	0.008	0.007	0.039	0.015	0.014
30123012	0.23	0.236	0.358	0.231	0.064	0.077	0.266
30126040	0.32	0.405	0.201	0.261	0.218	0.344	0.159
30162009	0.27	0.188	0.21	0.313	0.207	0.137	0.063
30162020	0.112	0.114	0.088	0.066	0.088	0.078	0.087
30162033	0.114	0.078	0.072	0.101	0.031	0.036	0.026

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4.3 Time series clustering

In this phase, we performed clustering of the generated time series data from the preprocessing phase.

4.3.1 Clustering

As mentioned in the literature review section, for clustering time series, two main decisions must be made: 1- choosing similarity measures such as Euclidean distance, COR, CORT, and DTW as described in the section 2; 2- selecting a clustering algorithm. According to what is mentioned in the section 2, we utilized a hierarchical clustering algorithm. In this study, four well-known similarity measures were selected. Using each similarity measure, clustering was conducted and results were evaluated in the next steps.

4.3.2 Selecting the best clustering model and visualizing the M attribute time series

Being an unsupervised technique, validation of the clustering results is a challenging task as there is no ground truth to perform comparisons. To choose the best clustering model, we set the number of clusters to 4, 5, 6, 7, and 8 and in each time, we employed the Ward clustering method using four similarity measures including Euclidean distance, COR, CORT, and DTW. The validity of the obtained clusters was tested using the silhouette index which is a conventional cluster validity index (Desgraupes, 2013). The value of silhouette can range between -1 and 1. For each data object in the resulted clusters, the silhouette value for that data point can be described as follows: if it is close to -1, then, the data object is clustered wrongly; and if it is close to 1, then, the data object is well clustered (Kalpakis et al., 2001). The results of clustering are summarized in Table 5. As explained, the higher value of silhouette indicates the better quality of the resulted clusters. That is the obtained clusters have high intra-cluster similarity and low inter-cluster similarity. As observed from Table 5, the highest silhouette value is achieved using the DTW measure with K=4 clusters. Therefore, the clustering model achieved using DTW with k=4 was selected as the final model for analysis purpose. In the next subsection, we analyze the customers' behavior using the model.

Table 5: The silhouette index for each combination of the cluster numbers and similarity measures (on M)

Similarity measure	K=4	K=5	K=6	K=7	K=8
Euclidean	0.23	0.18	0.19	0.17	0.18
COR	0.12	0.13	0.15	0.16	0.16
CORT	0.19	0.19	0.2	0.21	0.19
DTW	**0.35**	0.17	0.16	0.16	0.14

4.3.3 Selecting the best clustering model and visualizing the F attribute time series

Similar to the M time series, for the F time series, we conducted some sets of experiments in order to discover the best clustering models. More specifically, we performed clustering using combinations of different cluster numbers such as k=4, 5, 6, 7, 8 and different similarity measures involving Euclidean distance, COR, CORT, and DTW. Similar to the previous subsection, the Ward clustering algorithm was exploited for all the experiments. For each combination, the silhouette index is computed. The results are illustrated in Table 6. According to the table, the best clustering result is gained using DTW with k=4 as it obtains the highest silhouette value among the other models. Therefore, the model (DTW, K=4) is selected for comprehensive analysis in the next sections. The results of clustering on both the M and F attributes are in line with (Rakthanmanon et al., 2013, Ding et al., 2008) who concluded that DTW is the best similarity measures in the context of time series analysis.

Table 6: The silhouette index for each combination of the cluster numbers and similarity measures (on F)

Similarity measure	K=4	K=5	K=6	K=7	K=8
Euclidean	0.44	0.28	0.29	0.29	0.27
COR	0.14	0.11	0.11	0.12	0.12
CORT	0.37	0.32	0.25	0.25	0.25
DTW	**0.46**	0.44	0.31	0.29	0.28

4.4 Result analysis and managerial implications

In this step, we analyze the result of time series clustering on the M and F attributes. In order to illustrate the plots of Figure 2 and Figure 4 compactly, we plotted the mean trends of all clusters in a single figure to elucidate the analysis task (Figure 3 and Figure 5 for M and F, respectively). The analysis results and insights are given in the following subsections.

4.4.1 Analysis based on the trends of the monetary factor (M)

In this step, firstly, an appropriate label is assigned to each resulted customer group. Afterward, based on its trend, the overall status of that cluster is described and finally, corresponding marketing recommendation is proposed.

4.4.1.1 Labeling and describing customer groups and suggesting managerial implications

In this subsection, we have summarized the result of the analysis on the M factor and presented it in Table 7. In this table, for each cluster, according to its size and trend plot (Figure 2 and Figure 3), firstly, we label the cluster and then we briefly describe its characteristics and finally, possible marketing actions are suggested that can be pursued by the bank to enhance its relationship with those customer groups.

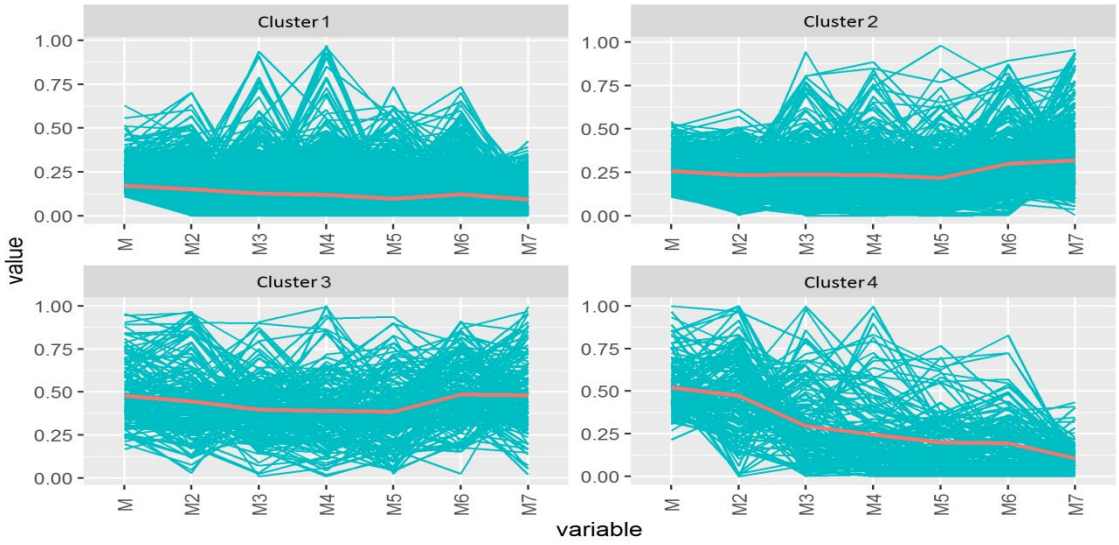


Figure 2: The best clustering results of time series with the mean for the M attribute

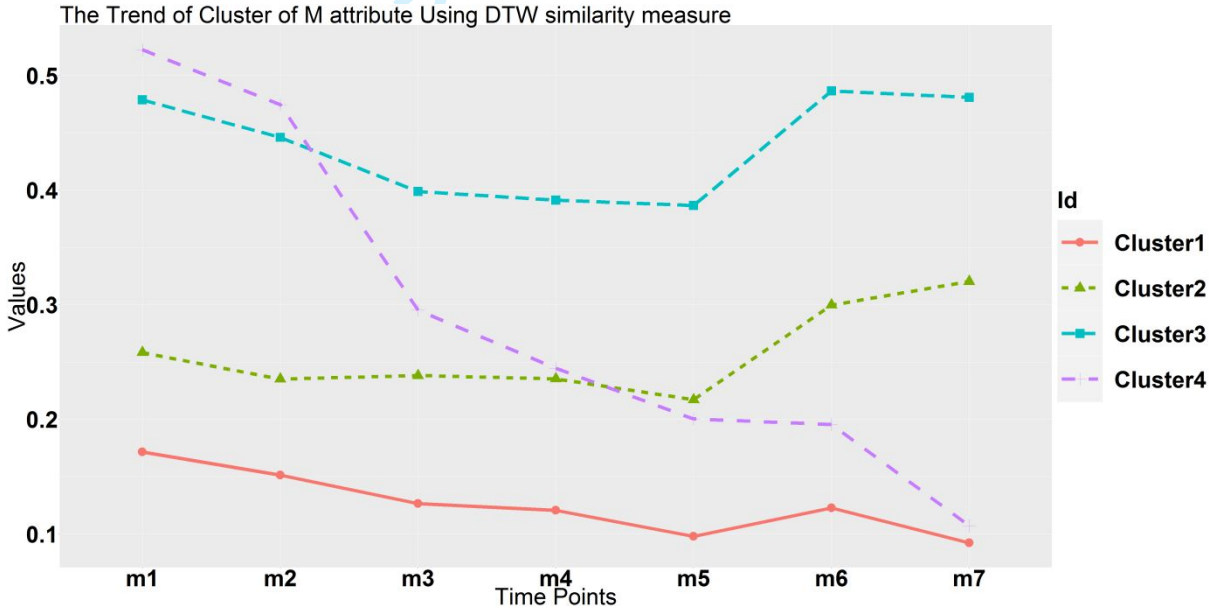


Figure 3: The mean trends of all the clusters for M

Table 7: Labeling customer groups and recommending managerial implication based on the results of M

Cluster	Size	Trend label	Description	Possible actions
Cluster1	1794	Prone to churn	This cluster which contains more than half of the customers demonstrates a disappointing	The bank should conduct interesting promotions to retain these customers and

			trend. In fact, these customers are going to be churned from the bank.	change them to high-value customers.
Cluster 2	418	Middle-value growing customers	The customers of this cluster have experienced a slight decrease in their value until the time point M5. However, the decrease is not significant and the cluster still is a high value cluster. After the M5 time point, this group of customers shows a positive trend and their values increase.	The positive trend initiated at M5 indicates that these customers have the potential to become more high-value customers. The bank should introduce more loyalty programs to engage the customers of this group to continue their growth in profitability.
Cluster 3	173	High-value growing customers	The trend of this cluster is similar to that of the cluster 2 with one difference that is the customers of this cluster have a higher value than those in the cluster 2. This customer group experiences a slight decline in their value until the time point M5 and then their value increases.	Similar to the cluster 2, the bank should take special actions to keep this cluster profitability. Moreover, this cluster is considered as the highest value cluster that has preserved its positive trend. Considering the fact that high-value customers produce a significant part of a firm revenue, it is necessary that the firm pursues an especial marketing program to keep this cluster and make it continue its growth. One important finding of analyzing M trends is that the bank should pay especial attention to its new customers at the early months. Exploring the trends of the cluster 2 and cluster 3 proves this evidence as both of these clusters experience a slight

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				decrease until the time point M5.
Cluster 4	146	Churners	As observed from the figure, at the beginning of the study, at the time point M1, the mean value of this cluster was the highest value. However, over the time, the customers of this cluster experience a steepest decrease in value and become the least valued cluster. In addition, it can be implied that the customers of this cluster have churned.	Although the number of customers of this cluster is small, the trend is very negative, and it is suggested that the bank makes a close observation on these customers to discover the main causes of such behavior. As the trend shows, the customers of this cluster may be churned. Customer churn is a big problem for many firms as emphasized in the literature (Abbasimehr et al., 2013, Fathian et al., 2016) and the bank should conduct effective programs to prevent customers from churning.

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The results of clustering and characteristics of the obtained clusters (as described in Table 7) indicate that most of the bank's customers including customers belonging to the clusters 1 and 4 were not successfully moderated and are going to churn or have the tendency to churn. In other words, it appears that customer retention programs of the bank are not successful. The bank should devise more effective retention and loyalty programs to prevent customers from churning. In addition, customers belonging to the clusters 2 and 3 have the overall positive trend in their behavior. These customers are valuable and the bank should devise customer development programs in order to expand customers' transaction frequency, transaction value and profitability.

Further analysis of the results indicates that for all the clusters, there is a decrease in values of the time point 5. This can be because of an annual event, the Ramadan month, at that time point. This lunar month may fall in any season of the year and affects consumers' routines such as purchasing.

4.4.2 Analysis based on the tends of the frequency (F) factor

Similar to the previous step, for each cluster firstly, cluster labeling is performed and then cluster characteristics are analyzed and finally, proper marketing suggestions are given.

4.4.2.1 Labeling and describing customer groups and suggesting managerial implications

The frequency factor of each customer denotes how many transactions each customer has performed. Moreover, it is obvious that the amount of money (monetary) of each transaction can

be any value. Therefore, the high number of transactions does not imply the high value of monetary. For example, some POS terminals have a small amount of M for each transaction, e.g. in the grocery. However, some POS terminals have a large amount of M per transaction, for example, consider an auto dealer company which is using POS. By investigating the trends of the F factor, we can measure the level of activity of a particular customer. As the value of F increases or decreases, the trend of customers becomes positive or negative, respectively. Therefore, in this section, we discover the trend of F by using time series analysis. As stated above, we make a relation or correspondence between the F trend and level of activity.

In this subsection, similar to the previous subsection, we have portrayed the result of analysis on the F factor and presented it in Table 8. In this table, for each cluster, according to the cluster size and its trend's plot (Figure 4), firstly, we label the cluster, and then we briefly describe its characteristics and finally, possible marketing actions are suggested that can be used by the bank to improve its relationship with that customer group.

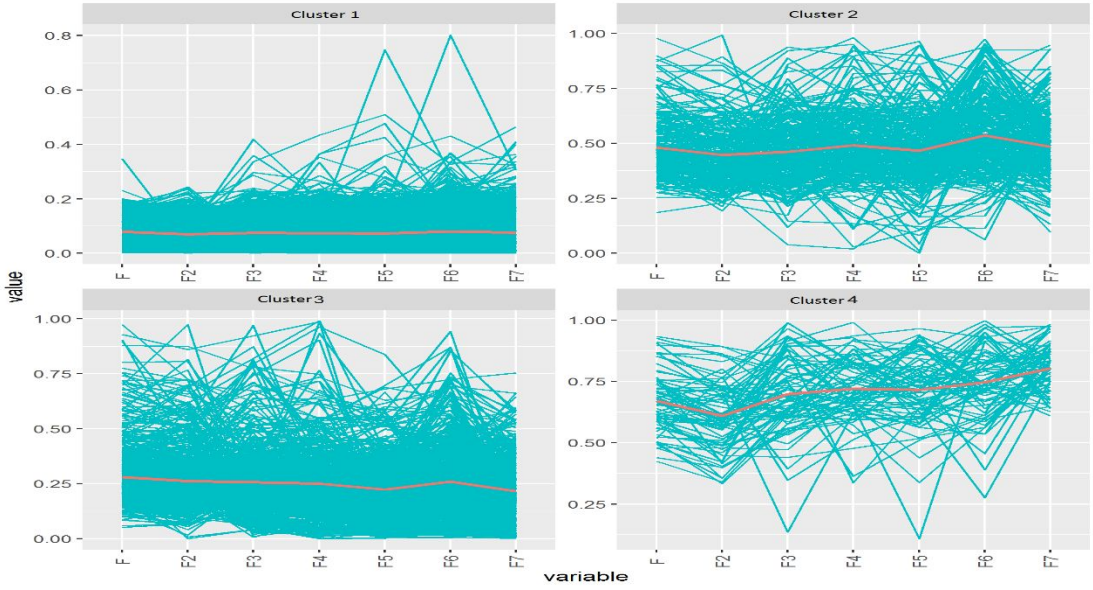


Figure 4: The best clustering results of time series with the mean for the F attribute

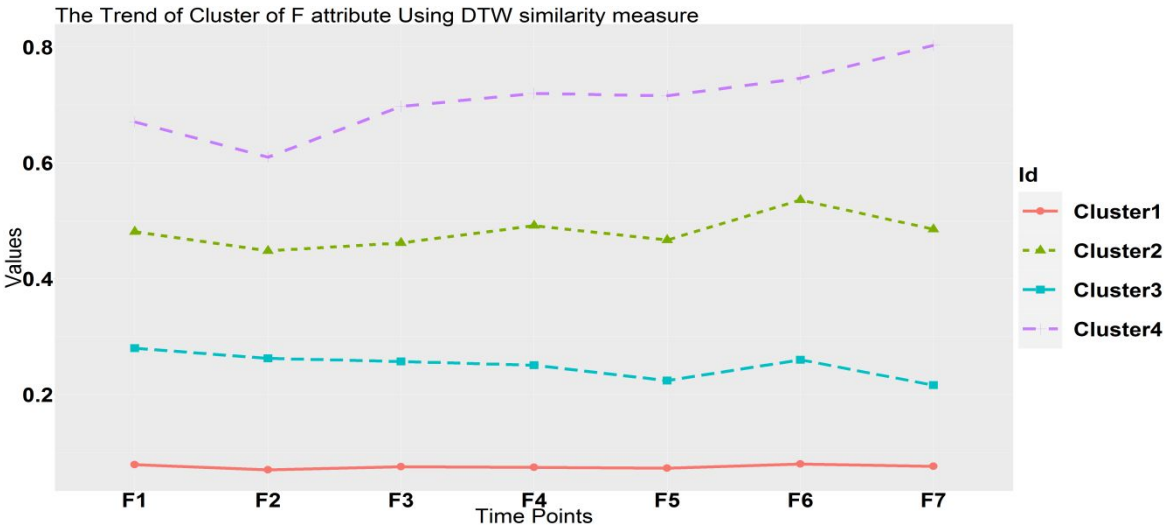


Figure 5: The mean trends for all the clusters for F

Table 8: Labeling customer groups and recommending managerial implication based on the results of F

Cluster	Size	Trend label	Description	Possible actions
Cluster 1	1423	Least active customers	This is a least active customer group that preserves its stability over time.	Although a high F does not indicate a high M, the trend of this cluster gives important clues about the activity of customers over time. As the size of this cluster is significant, the bank should strengthen its relationship with this cluster in order to increase its market share.
Cluster 2	275	Medium activity cluster with the positive trend	As observed from the figure, this cluster has a fluctuated but positive behavior. Except the time point 2 in which there is a decrease in the value of F, at the next time points, this cluster has a fluctuated behavior but positive trend. The reason for this fluctuation should be investigated.	Analyzing the behavior of this customer group in terms of F is a challenging task as this group shows a fluctuated trend. However, the trend is positive after the time point 2. One reason for this fluctuation may be the differences in the nature of the industry which the customer firm operates. Some industries demonstrate the seasonal behavior. Another probable reason may be the effect of marketing programs of rival banks. A thorough investigation should be accomplished in order to discover the reason for this fluctuation.

Cluster 3	756	low-level with the roughly stable trend	This customer group experiences a minor decrease until the time point 5 and then returns back to its initial state in the time point 6, but again shows a slight decrease in the last time point. The important fact about this cluster is that the fluctuation is not considerable and the behavior is approximately stable.	Considering the size of the cluster and its performance in terms of F, the bank should persuade this group to increase their activity and show a positive trend.
Cluster 4	77	Most active customers with the positive trend	This cluster contains high-level customers that have continued their growth in terms of F. Although, there is a decrease in the time point 2, its positive trend continues over time.	As this is the high-level customer group in terms of F, the bank should conduct especial loyalty programs to preserve the activeness of this group. Although, there is no clear relationship between F (frequency of transaction) and M (monetary), the increase in activity level for a given customer may lead to an increase in M over time.

As previously stated, a high F does not lead to a high M; for example, the monetary value per transaction is different for customers that are active in different guilds (e.g., the grocery guild against the car dealer guild). However, we consider the F attribute of a customer as its level of activity over the period of analysis. The results of time series clustering based on F indicate that customers belonging to the clusters 1 and 3 experience a decrease in F. Thus, the bank should strengthen its relationship with these clusters to increase its market share.

5. Conclusion

Customer segmentation is a crucial CRM practice which enables businesses to achieve a deeper understanding of customers' needs and behaviors. In this paper, we have proposed a new methodology for customer segmentation by involving customer behavior changes over time in the analysis using time series clustering. In fact, the main advantage of the proposed methodology is its ability to analyze customers by taking into account the evolving behavior of them. This methodology is successfully employed on a bank's transaction dataset over a duration of seven months. In order to extract meaningful groups of customer trends, different similarity measures

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with hierarchical clustering were evaluated. The obtained distinctive trends of the M and F attributes illustrate dominant behavioral customer patterns which are useful for the target bank's customer management strategies and can lead to efficient marketing plans. Based on our empirical study, we realized that the churner customer group and ascending growth customer groups constitute a significant portion of the customer base. Therefore, it is strongly suggested to banks to conduct special marketing strategies to prevent customers from churning. Discovering customer behavior trends over time rather than at a single time point indicates the effectiveness of the proposed methodology.

6. Limitations and future research

In this study, we only accessed the data of a seven-month time period. It would be preferred to analyze the data of a 12-month time period; however, we had some limitation in accessing data.

For our future research, we will implement time series clustering with some other distance measures including OWA-based distance measures, as proposed previously (Merigó et al., 2017, Merigo and Yager, 2013, Merigó and Gil-Lafuente, 2008). As another future work, we will combine the result of the proposed methodology and the RFM model to introduce a new customer segmentation model. We believe that the new model will eliminate the shortcoming of the conventional RFM model in dealing with customer dynamics.

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Dear Prof. Gandolfo Dominici,
Hello

We thank you and reviewer for studying our manuscript entitled " A new methodology for customer behavior analysis using time series clustering: a case study on a bank's customers" and providing the valuable comments. We have made all the changes reviewer suggested in your letter and addressed the comments of the reviewer in the notes below. Please note that our responses are shown in blue. Also please note that we have highlighted the modified parts of the paper with yellow color in the revised manuscript.

Best Regards,
Dr Hossein Abbasimehr
Assistant Professor
IT Group, Department of Information Technology and Computer Engineering
Azarbaijan Shahid Madani University
Tabriz, Iran

Reviewers' Comments to Author:
Reviewer: 1
Comments:
It is an interesting methodology and seems it could be tested in different industries and environments which makes it valuable. For segmentation purposes, it could help both researchers and practitioners in future jobs. Please make sure to fix the 3 sentences stated in point 6.

Thank you for suggestion; according to your comment, we have corrected the three sentences you have mentioned and also we have checked the manuscript to correct grammatical mistakes.

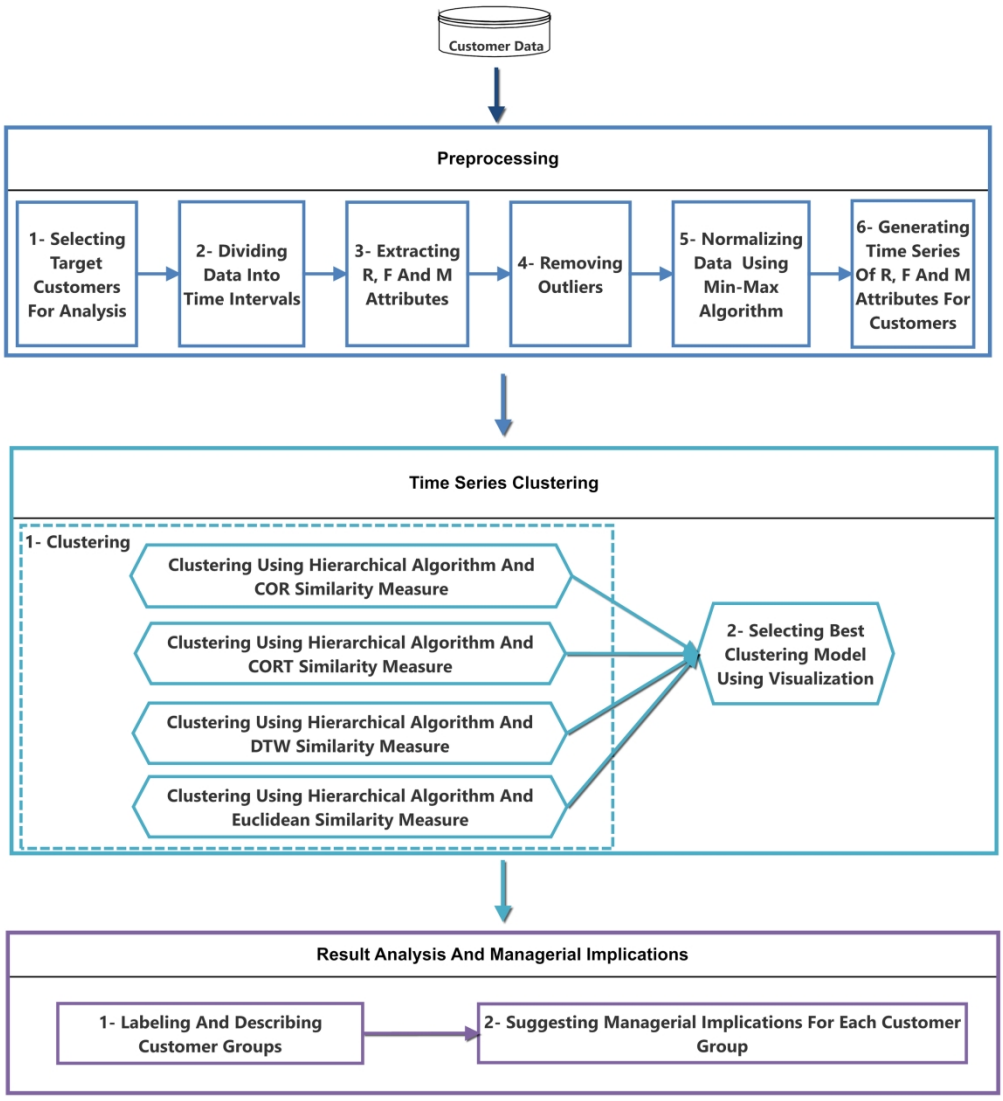


Figure 1: The steps of the proposed methodology

424x463mm (150 x 150 DPI)

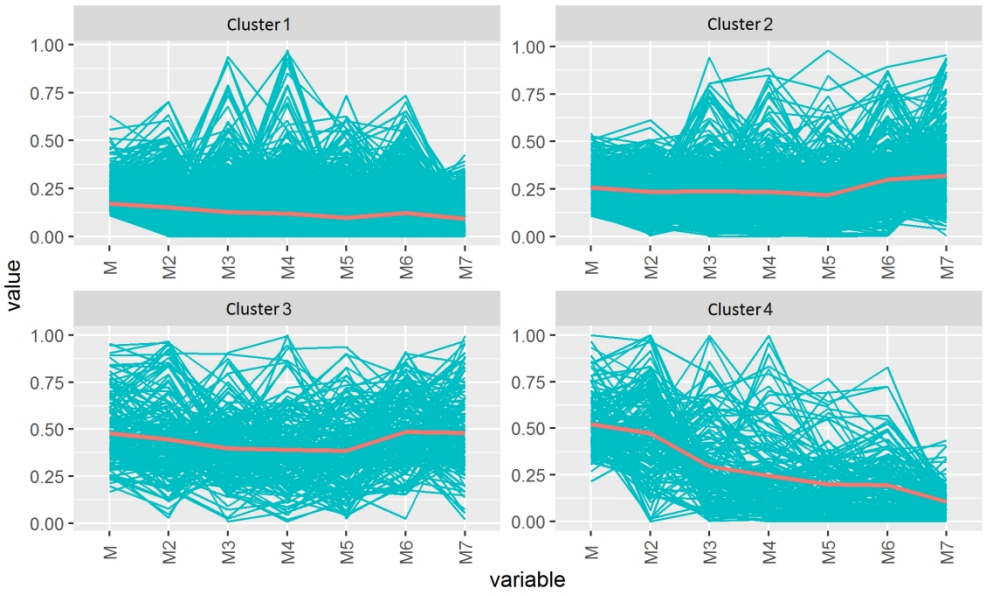


Figure 2: The best clustering results of time series with Mean for M attribute

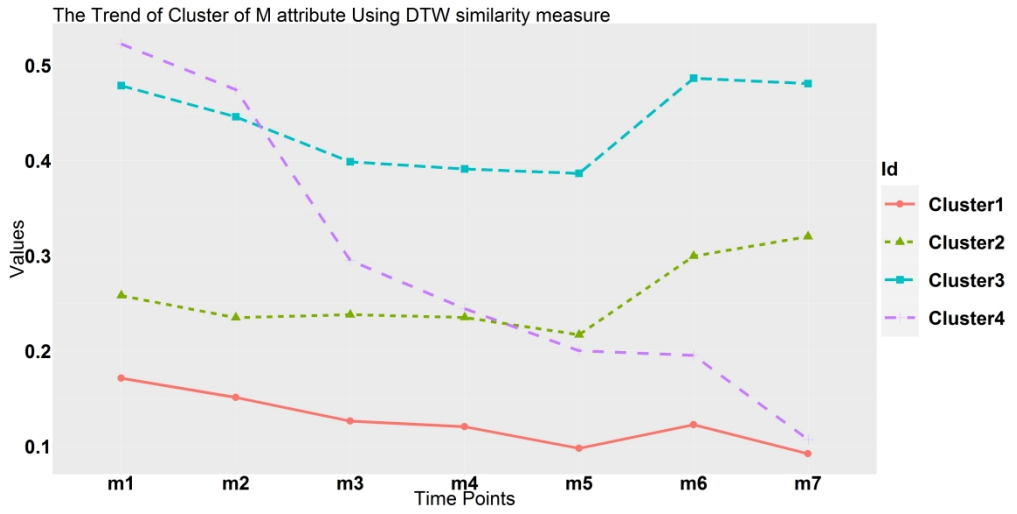


Figure 3: Mean trends of all clusters for M

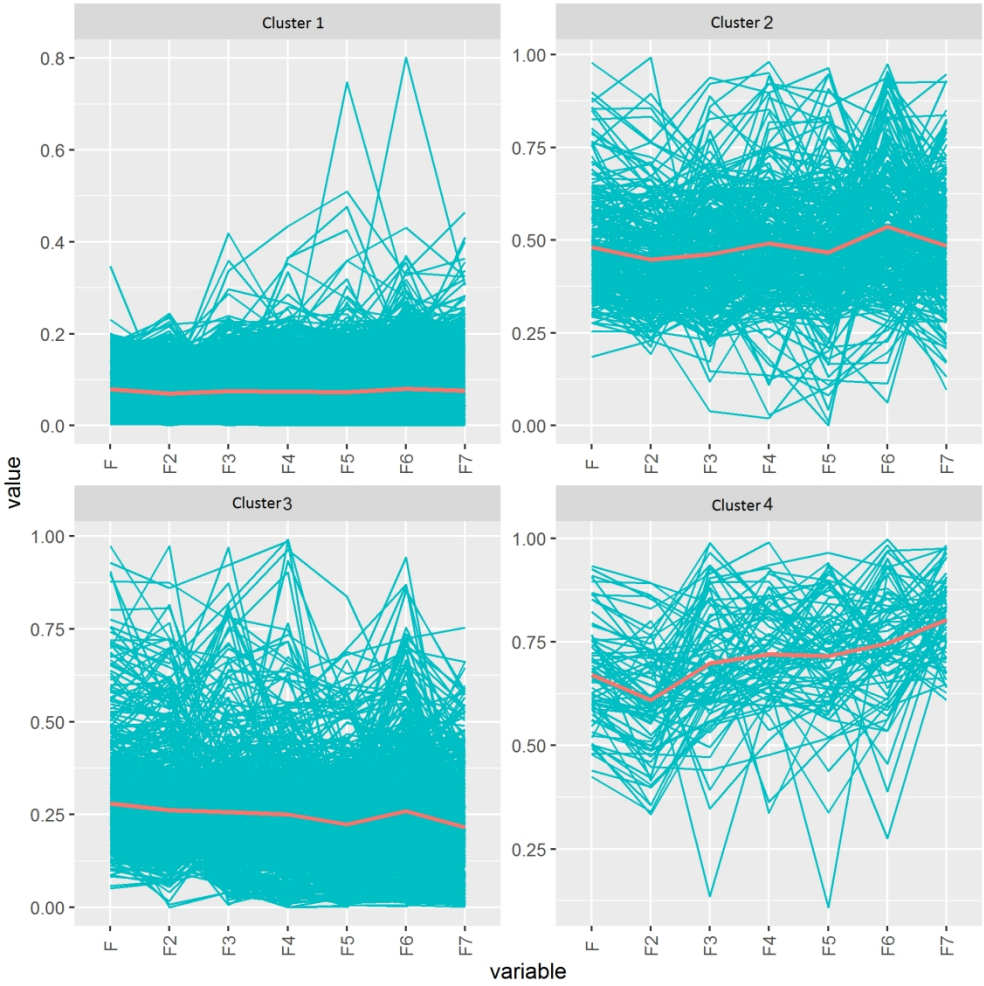


Figure 4: The best clustering results of time series with Mean for F attribute

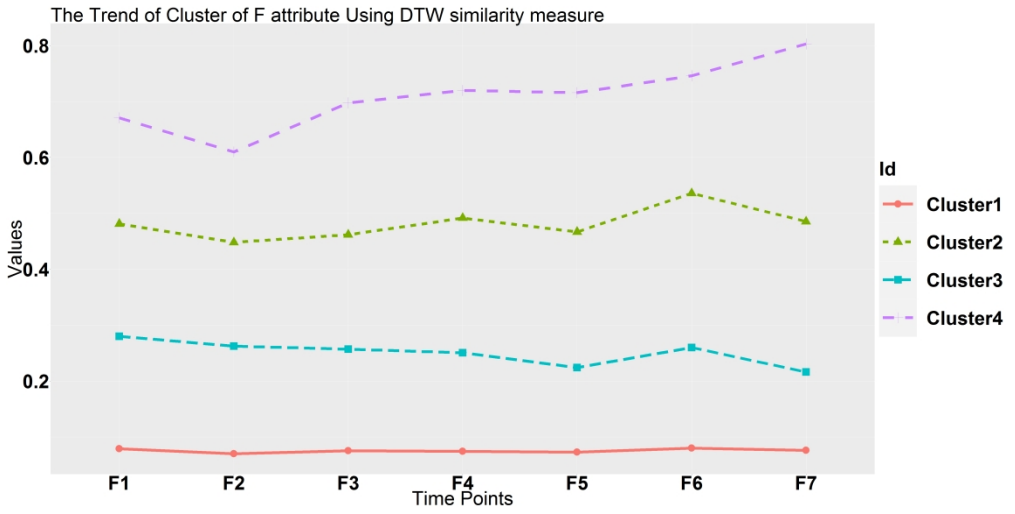


Figure 5: Mean trends for all clusters for F